StreamScale

Kafka → transactional PostgreSQL · scaling experiment

27 measured runs · 3 modes × 3 scenarios × 3 repetitions

All acknowledged IDs reconciled before/after restart

Same image, logical scenario sequence, resources and partitions

30-second warmup + 60-second base window; late fixtures extend the tail

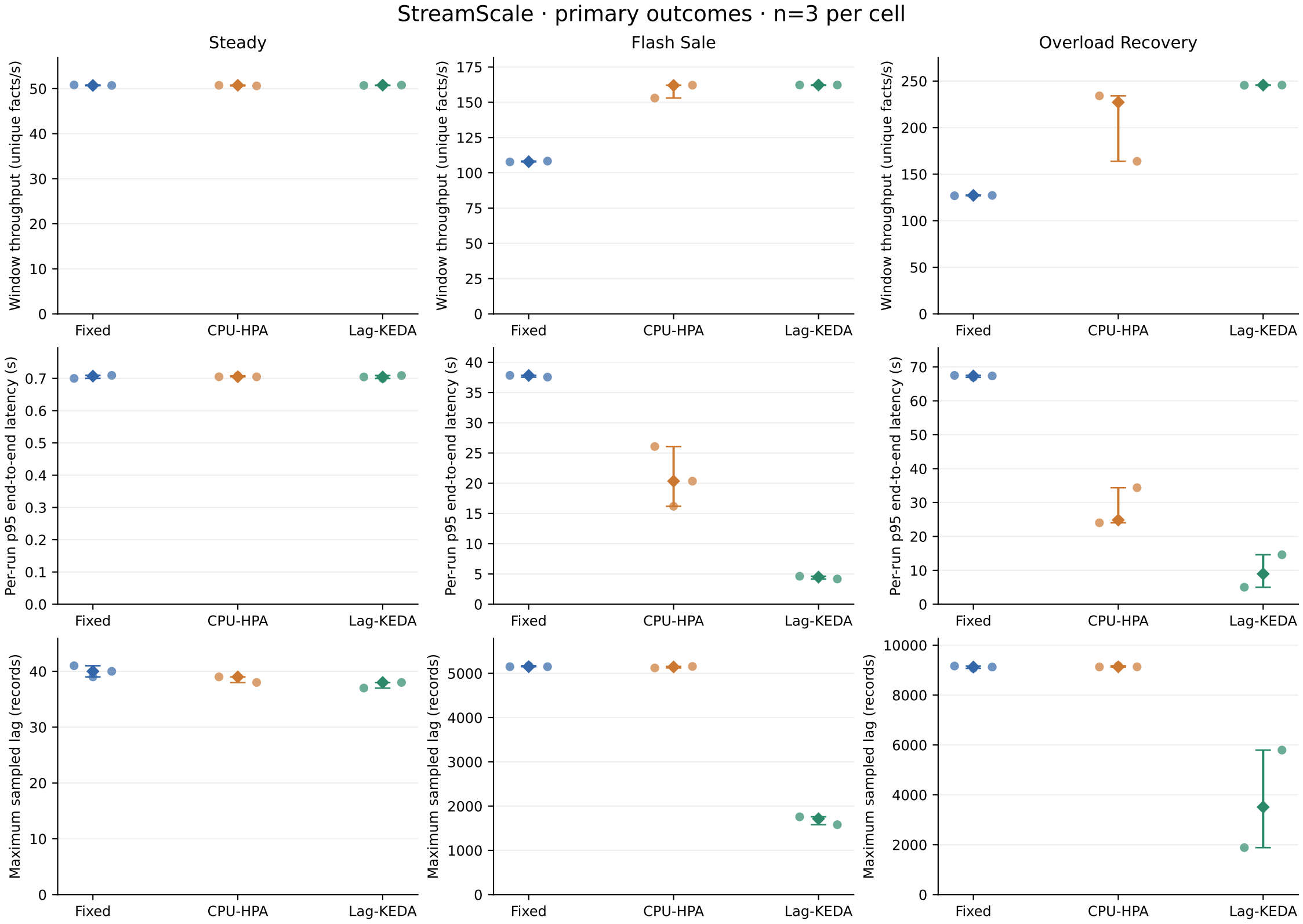
Frozen rates: 52/s steady · 390/s surge

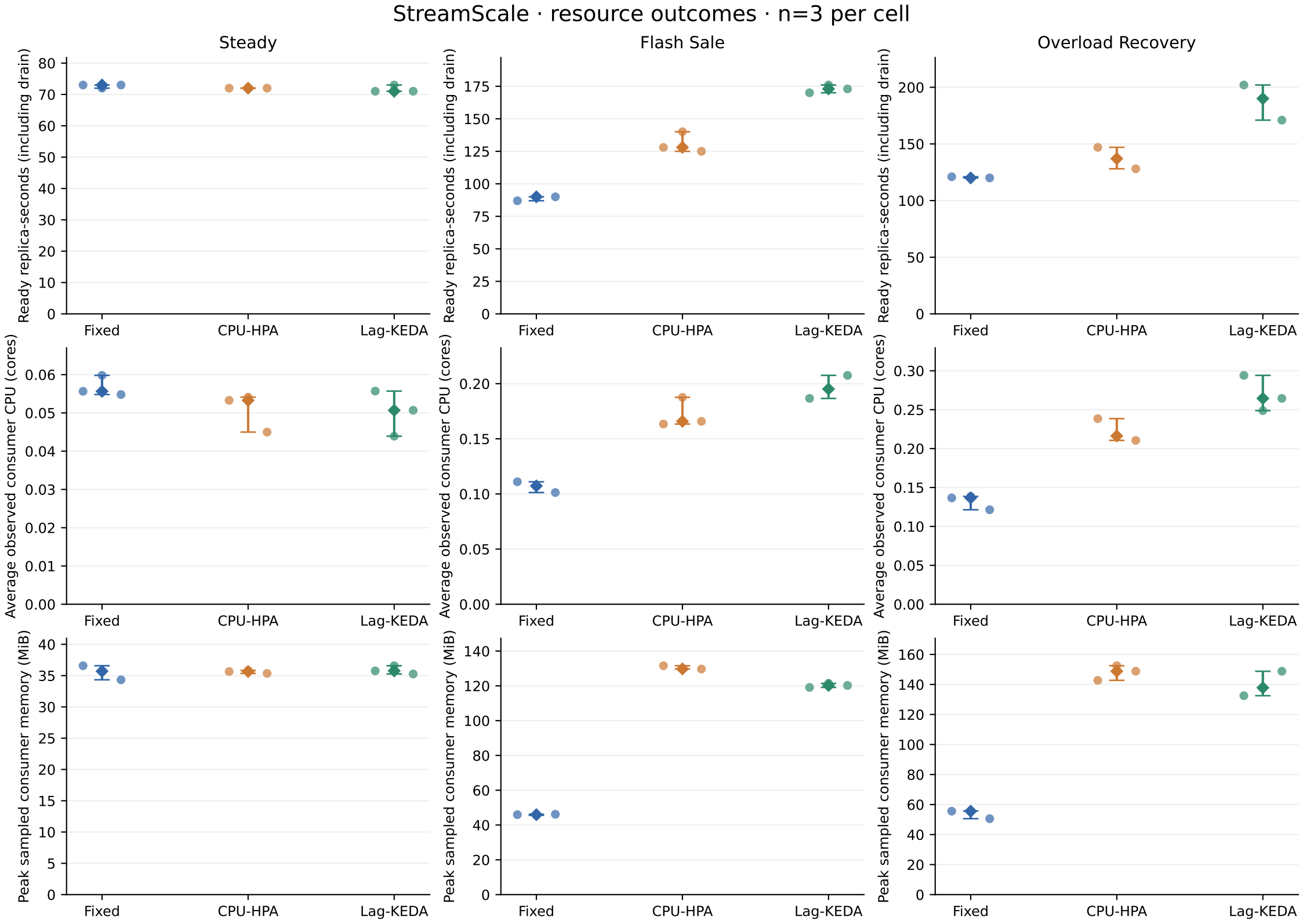
Dots = repetitions; diamonds = median; whiskers = observed range

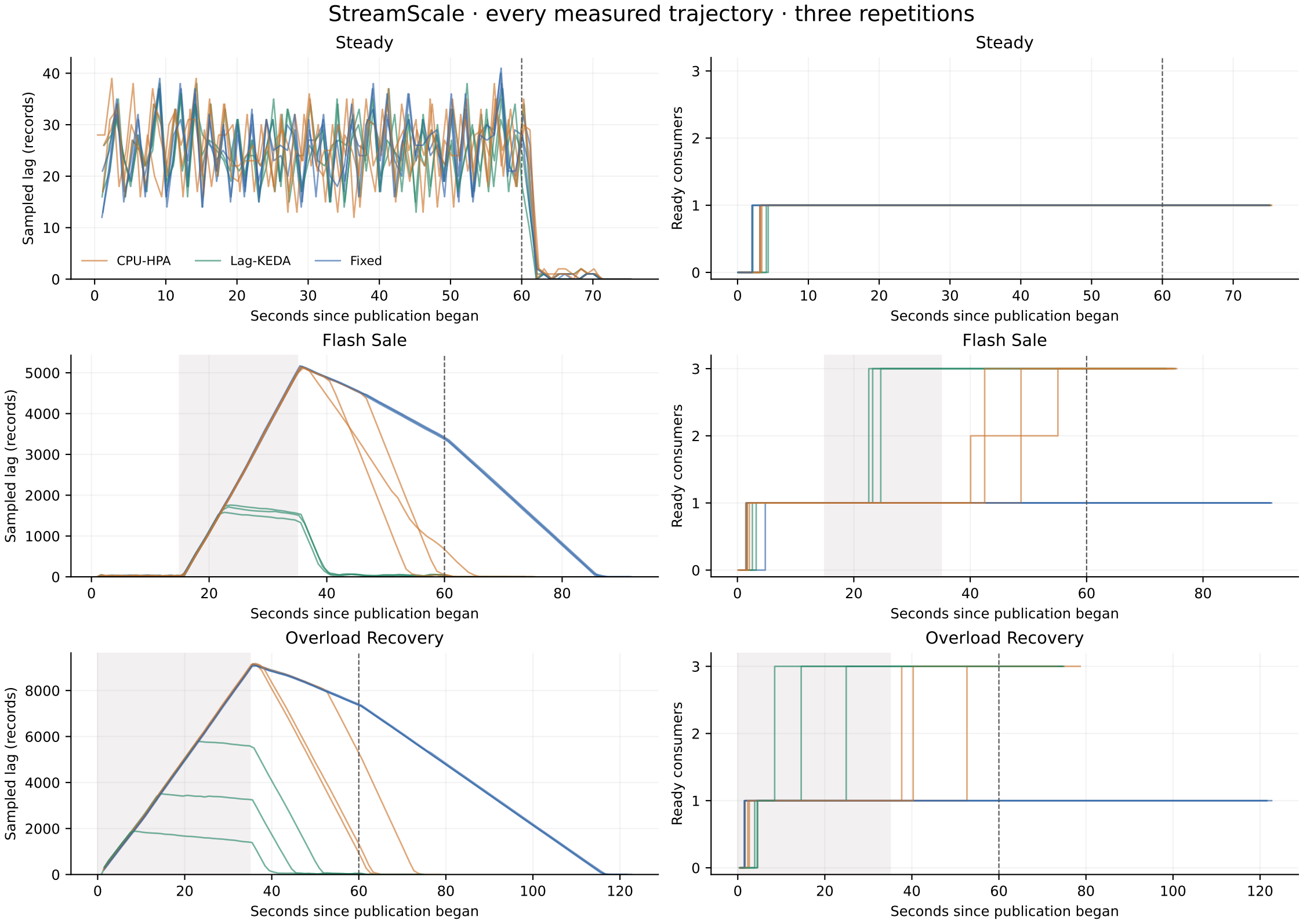
SQL p95 includes the drained tail; throughput uses the publishing window

One shared kind node, small sample, sampled resources; descriptive results

See report.md, summary.csv, frozen-protocol.json and provenance.json


StreamScale · primary outcomes · n=3 per cell
Steady
Flash Sale
Overload Recovery
Window throughput (unique facts/s)
Per-run p95 end-to-end latency (s)
Maximum sampled lag (records)
Fixed
CPU-HPA
Lag-KEDA


StreamScale · resource outcomes · n=3 per cell
Steady
Flash Sale
Overload Recovery
Ready replica-seconds (including drain)
Average observed consumer CPU (cores)
Peak sampled consumer memory (MiB)
Fixed
CPU-HPA
Lag-KEDA


StreamScale · every measured trajectory · three repetitions
Steady
Steady
Sampled lag (records)
Ready consumers
CPU-HPA
Lag-KEDA
Fixed
Seconds since publication began
Seconds since publication began
Flash Sale
Flash Sale
Sampled lag (records)
Ready consumers
Seconds since publication began
Seconds since publication began
Overload Recovery
Overload Recovery
Sampled lag (records)
Ready consumers
Seconds since publication began
Seconds since publication began